

JOHANNES SPÄTH, PhD

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Computer scientist and engineer with 12 years of experience spanning academic research, startup founding, and technical leadership. Specializes in program analysis, formal verification, and software security, with a track record of translating deep theoretical work into production tools and commercial products.

WORK EXPERIENCE (12 YEARS)

Certora Ltd

August 2023 - July 2026

R&D Director (since December 2023, initially hired as Senior Software Engineer) Munich, Germany

Certora provides security audits and automated formal verification for web3 protocols. With the “Prover”, Certora builds a product that uses SMT solvers to mathematically verify the implementation of smart contracts.

Responsibilities & Tasks

- Leading a team of up to seven compiler and software engineers to enhance the Prover across different blockchain ecosystems (Ethereum, Stellar, Solana)
- Contributed to the Prover and to Certora’s specification language (CVL and CVLR): added support for strong invariant logic, lifted LLVM debug information to present counterexamples in IDEs, implemented a program interpreter over the IR, and worked on LLM-based spec generation
- Delivered hands-on Certora Prover workshops at TrustX 2023 (Istanbul) and Meridian 2024 (London), introducing 20+ developers to formal verification techniques for smart contract security

CodeShield GmbH

November 2019 - June 2023

CEO & Co-Founder Paderborn / Munich, Germany

CodeShield is a Software-as-a-Service that detects and fixes cloud security vulnerabilities (Gartner terms: CIEM, CNAPP, CSPM).

Responsibilities & Tasks

- Acquiring and implementing proof-of-concepts with national and international clients
- Securing 1.2M EUR in funding to build and grow the team
- Leading customer research, defining the product vision, iterating on MVP: Discover IAM permission misconfigurations and demonstrate exploitable attack vectors (IAM privilege escalations)

Fraunhofer IEM - Institute for Mechatronic Systems Design

March 2017 - October 2019

Senior Expert & Research Associate Paderborn, Germany

The department of Software Engineering and IT Security at Fraunhofer IEM develops and researches on methods and tools for secure development, maintenance, and operation of software-intensive systems.

Responsibilities & Tasks

- Research on efficient and precise program analysis with focus on security (pointer and taint analyses)
- Development of the IDE plugin Eclipse CogniCrypt: a spell checker for developers detecting insecure usages of cryptographic APIs, presented at EclipseCon Europe 2018 and heise devSec 2019
- Integration of static code analyses with industry partners (SAST & SCA for DevOps)
- Acquisition of industry and research projects with volume of more than 1M EUR

Oracle

Research Intern

August 2016 - January 2017

Brisbane, Australia

Oracle Labs Brisbane is one of Oracle's eight research centers. The team in Brisbane develops program analysis technologies for internal use (Oracle Parfait).

Responsibilities & Tasks

- Research stay to work on existing data flow and code analyses solution
- Prototyping and implementing concepts for efficient program analysis based on soufflé (Datalog-based program analysis)
- C++-implementation for points-to summaries on top of soufflé

Fraunhofer SIT - Institute for Secure Information Technology

Research Associate

March 2014 - July 2016

Darmstadt, Germany

Responsibilities & Tasks

- Researching and prototyping program and data-flow analyses (pointer and taint analyses)
- Security analysis of the source code of TrueCrypt using Coverity Scan and Fortify, commissioned by the German Federal Office for Information Security (BSI)
- Security & pen testing of web applications using OWASP ZAP & Wireshark

ACADEMIC DEGREES

PhD in Computer Science (Dr. rer. nat.)

January 2019

Paderborn University

Dissertation: *Synchronized Pushdown Systems for Pointer and Data-Flow Analysis*

Grade: *summa cum laude* (received 3 awards)

Master of Science in Mathematics with Specialisation in Computer Science

September 2013

Technical University of Darmstadt

Grade: 1.6

Bachelor of Science in Mathematics

March 2011

Johannes Gutenberg University of Mainz

Grade: 1.6

TECHNOLOGY SKILLS

- Programming languages: Kotlin / Java, TypeScript / JavaScript, Rust, Python, C++
- Infrastructure & build systems: AWS, docker, maven, npm, git, cargo, uv
- Program-analysis and formal-verification-related: Soot, WALA, Datalog, soufflé, SMT solving, LLVM with bytecode targets: SBF for Solana and WASM for Stellar

SPOKEN LANGUAGES

- German - native
- English - fluent

SELECTED PUBLICATIONS & PATENTS

- *CrySL: An Extensible Approach to Validating the Correct Usage of Cryptographic APIs.*
Stefan Krüger, Johannes Späth, Karim Ali, Eric Bodden, Mira Mezini. *In Transactions on Software Engineering (TSE)*, 2019.

- *Context-, Flow- and Field- Sensitive Data-Flow Analysis using Synchronized Pushdown Systems.*
Johannes Späth, Karim Ali, Eric Bodden. *In Symposium on Principles of Programming Languages (POPL), 2019.*
- *Boomerang: Demand-Driven Flow- and Context-Sensitive Pointer Analysis for Java.*
Johannes Späth, Lisa Nguyen Quang Do, Karim Ali, Eric Bodden. *In European Conference on Object-Oriented Programming (ECOOP), 2016.*
- *Security Analysis of TrueCrypt*
Mauro Baluda, Andreas Fuchs, Philipp Holzinger, Lisa Nguyen, Lotfi ben Othmane, Andreas Poller, Jürgen Repp, Johannes Späth, Jan Steffan, Stefan Triller, Eric Bodden. *Federal Office for Information Security, 2015.*
- *Modular Points-To Analysis. (Patent, Publication Number: 20190065180)*
Johannes Späth, Francois Gauthier, Padmanabhan Krishnan. *Oracle International Corporation, 2017.*

AWARDS

- *Fraunhofer ICT Dissertation Award 2022*
- *Award by the Presidium of the University of Paderborn (Excellent Dissertation 2019)*
- *Ernst Denert Software-Engineering Award 2019*
- *Distinguished Paper Award (POPL 2019)*
- *Artifact Evaluation Award (POPL 2019, ECOOP 2016)*